CSCI 5526

Convolutional Neural Networks for Medical Imaging and Time-series

Announcements

- MP3 was released!
- Final project groups must be finalized by Friday!

Student Questions

Has PyTorch basically taken over as the main deep learning library at this point? Tensorflow used to be quite popular but hasn't heard much about it recently.

-- Yes, especially for research and development.

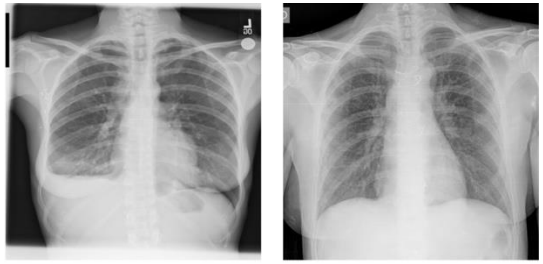
-- But Tensorflow is still used by certain companies, mainly in production environments.

Outline

- ML tasks related to medical image and time-series
- History of medical image and time-series feature extraction
- Convolutions as universal feature extractors
- Convolutional neural networks
- Some specific architectures
  - U-net
  - ECG architectures
  - EEG architectures

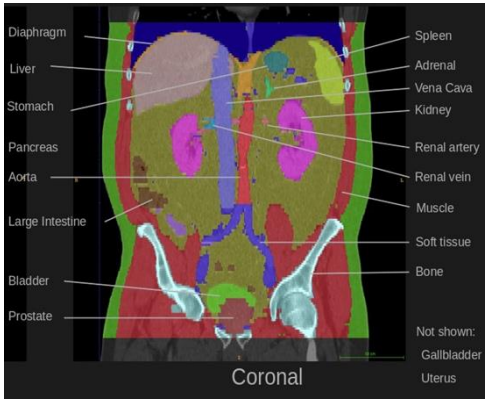
Medical Imaging ML Tasks

Classification

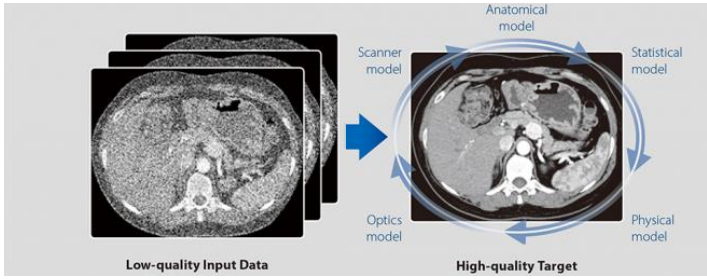


(b) Pneumonia (c) COVID-19

Segmentation



Reconstruction

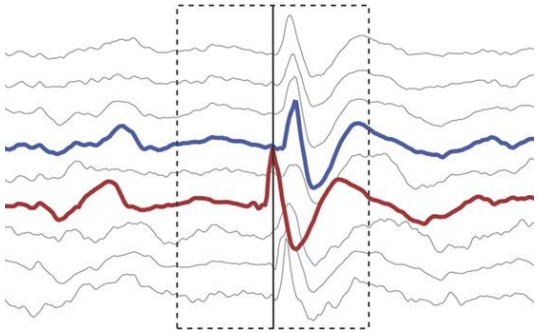


Medical Timeseries ML Tasks

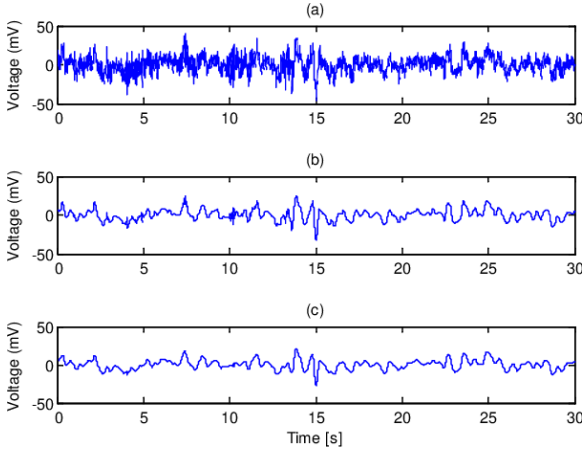
Rhythm Classification



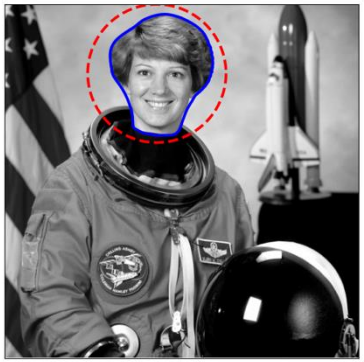
Detection

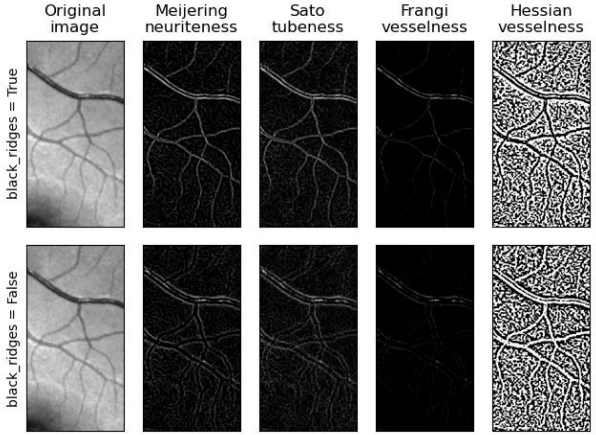


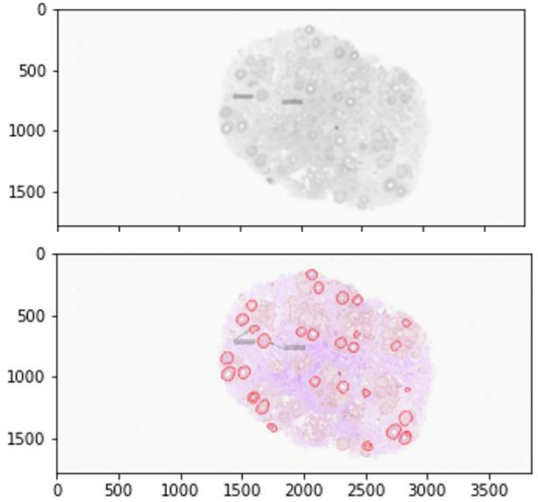
Denoising



Traditional Computer Vision

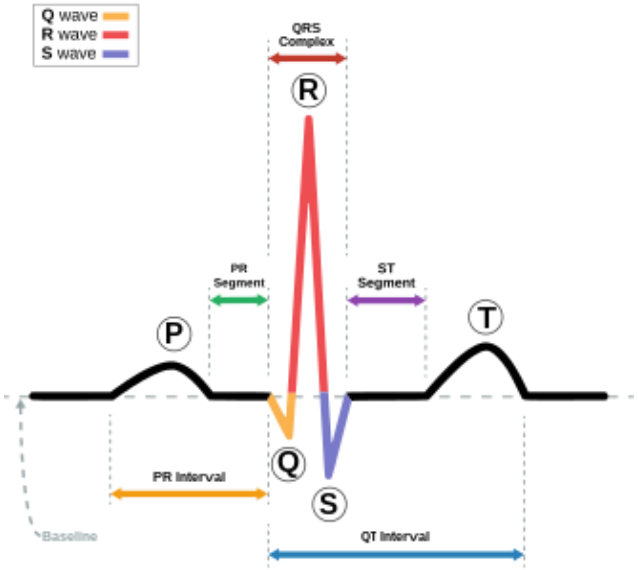


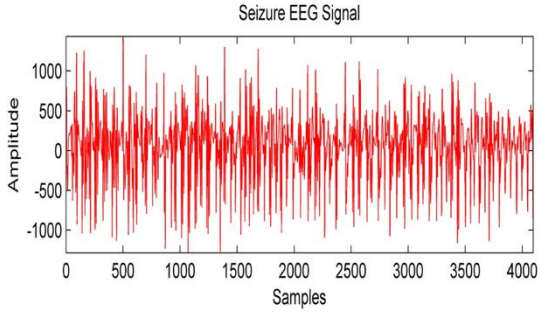


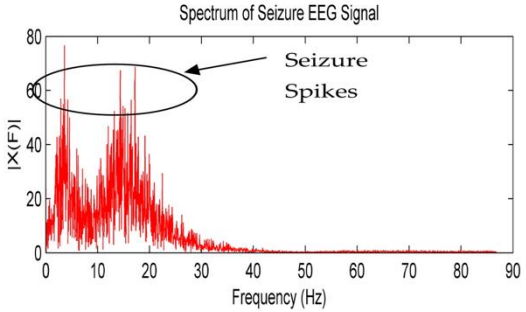


Examples from scikit-image.org

Traditional Timeseries Feature Engineering







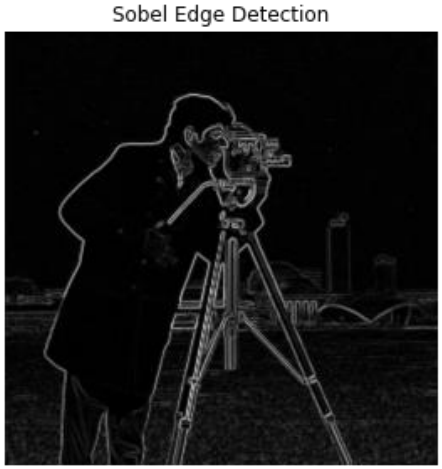
Convolutions as Universal Feature Extractors

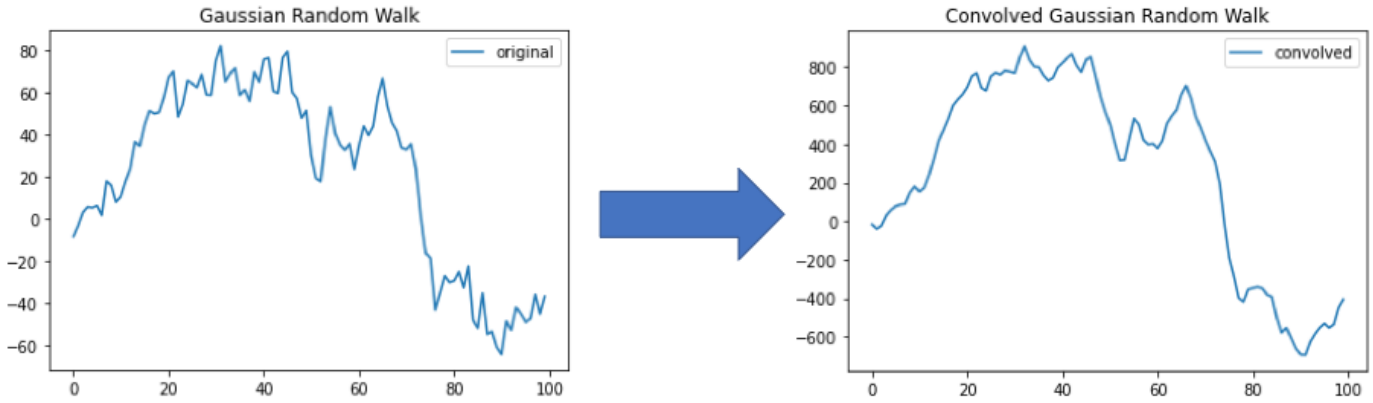
- An array operation where each output data element is a weighted sum of a collection of neighboring input elements

- The weights used in the weighted sum calculation are defined by an input mask array, commonly referred to as the *convolution kernel*
  - Sometimes referred to as convolution masks or convolution filters.
  - The same kernel is typically used for all elements of the array.

Some Examples







Convolution Mathematics

$$f(x) * g(x) = \int_{-\infty}^{\infty} f(\tau).g(\tau - x)d\tau$$

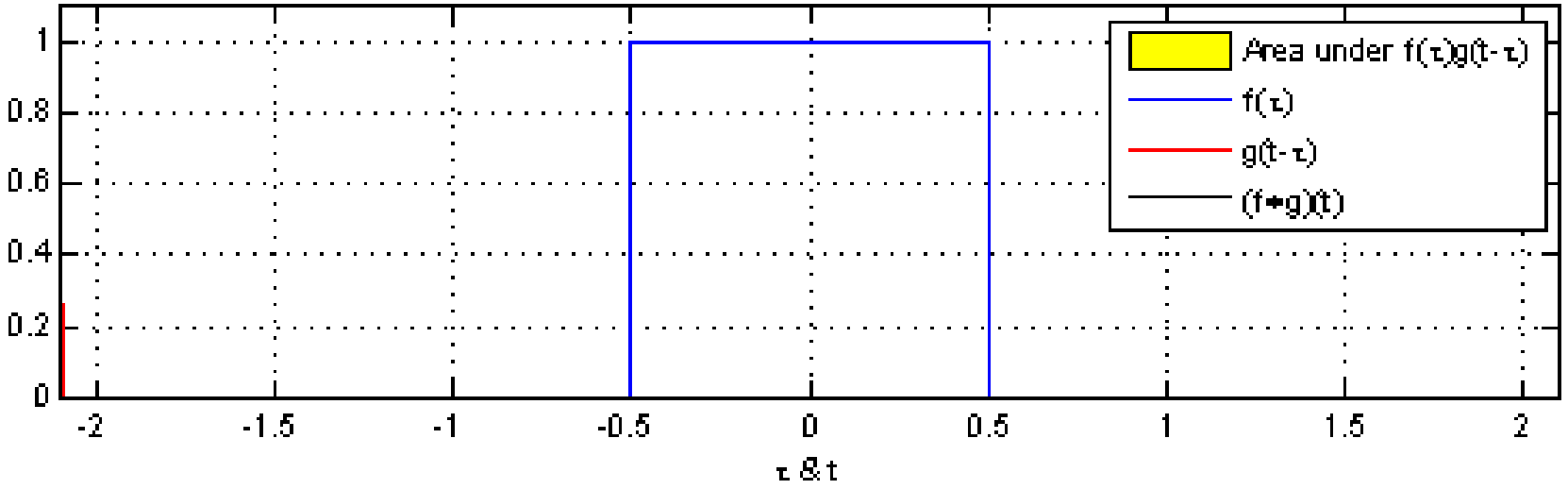
$$f[x] * g[x] = \sum_{k=-\infty}^{\infty} f[k].g[k - x]$$

1. Flip: $g(x) \rightarrow g(-x)$
2. Shift: $g(\tau - x)$
3. Compute area overlap: $f(\tau).g(\tau - x)$

4. Convolution is commutative:

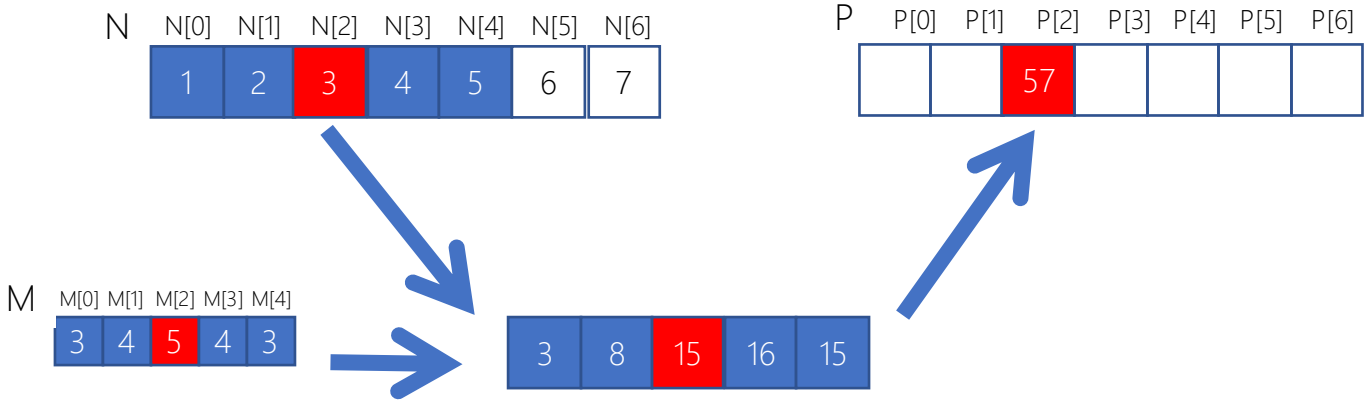
$$f(x) * g(x) = g(x) * f(x)$$

Example



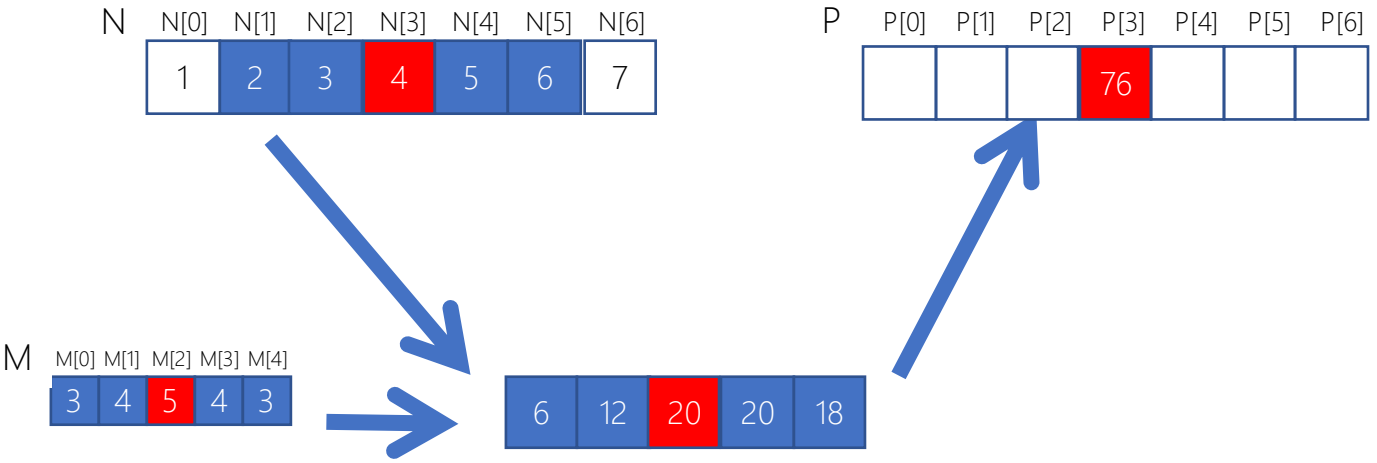
1D Convolution Example

- Commonly used for signal processing
  - MASK_WIDTH is usually an odd number of elements for symmetry (5 in this example)
  - MASK_RADIUS is the number of elements used in convolution on each side of the pixel being calculated (2 in this example).
- Calculation of P[2]:



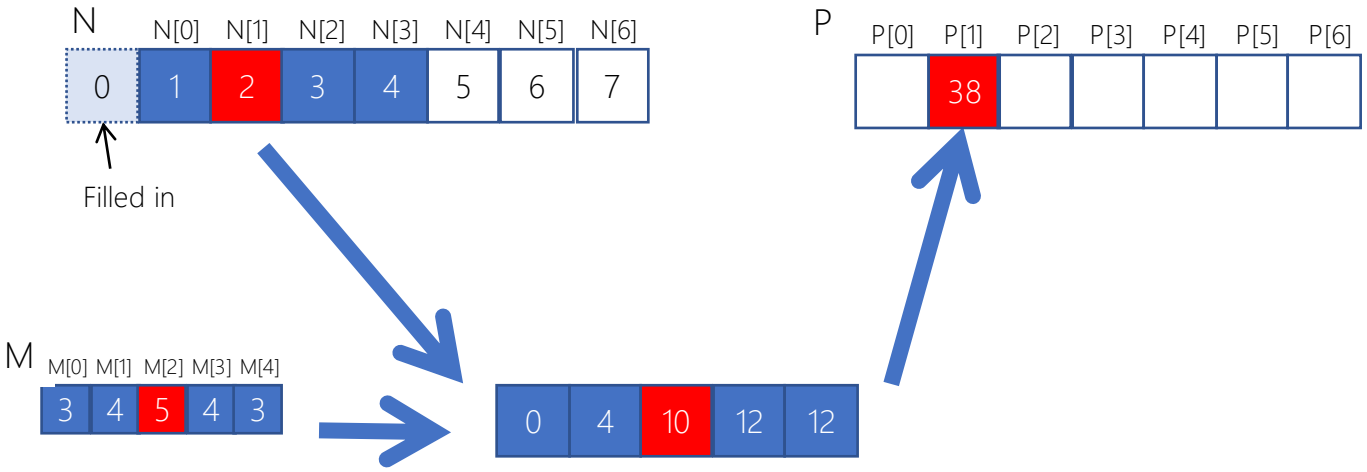
1D Convolution Example

- Calculation of P[3]



1D Convolution Boundaries

- Calculation of output elements near the boundaries of the input array need to deal with "ghost" elements
  - Different policies (0, replicates of boundary values, etc.)



Padding

- The method by which we perform convolution around array boundaries.
  - Typically involves adding extra elements.

- Different ways
  - Full padding
  - Valid padding
  - Same padding

Full Padding

```
Input              :  [5  6  7  8  9]
Filter             :  [ 1   0 -1]
Filter flipped     :  [ -1   0  1]
Full convolution:  [ 5   6   2   2   2 -8 -9]
```

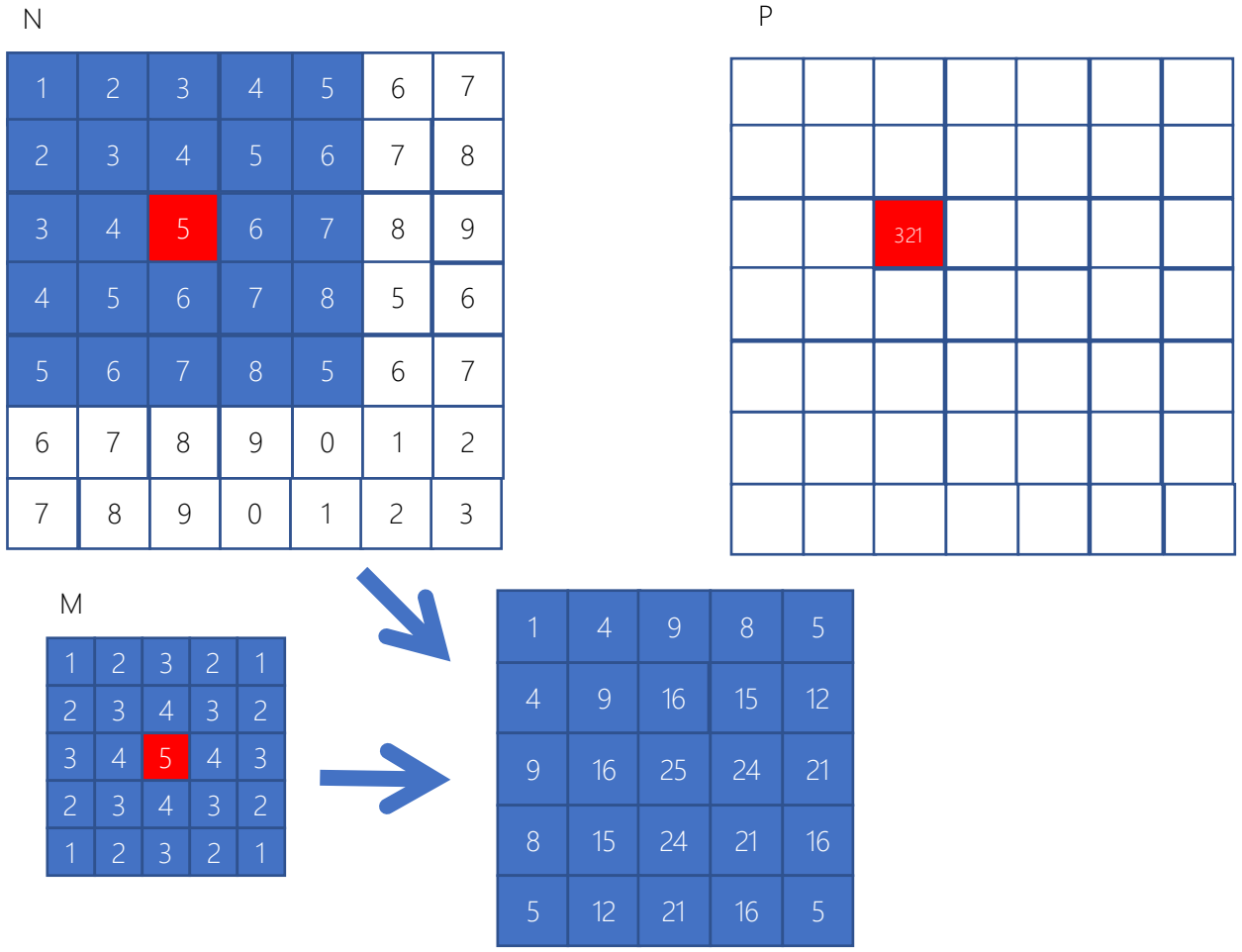
Valid Padding

```
Input              : [5 6 7 8 9]
Filter             : [ 1 0 -1]
Filter flipped     : [ -1 0 1]
Valid convolution  : [2 2 2]
```

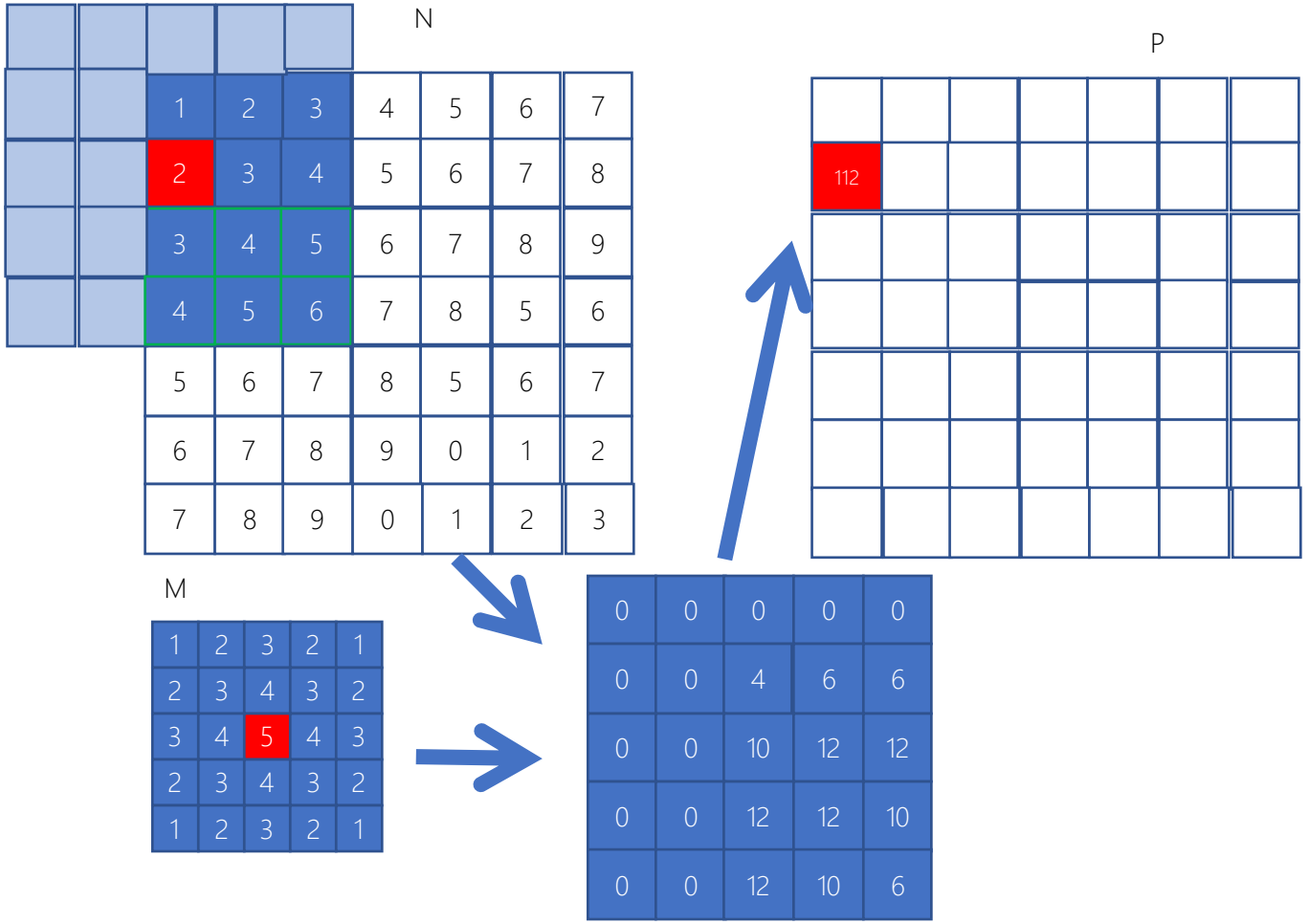
Same Padding

```
Input              :    [5 6 7 8 9]
Filter             :    [ 1  0 -1]
Filter flipped     :    [ -1 0 1]
Same convolution   :    [ 6  2  2  2 -8]
```

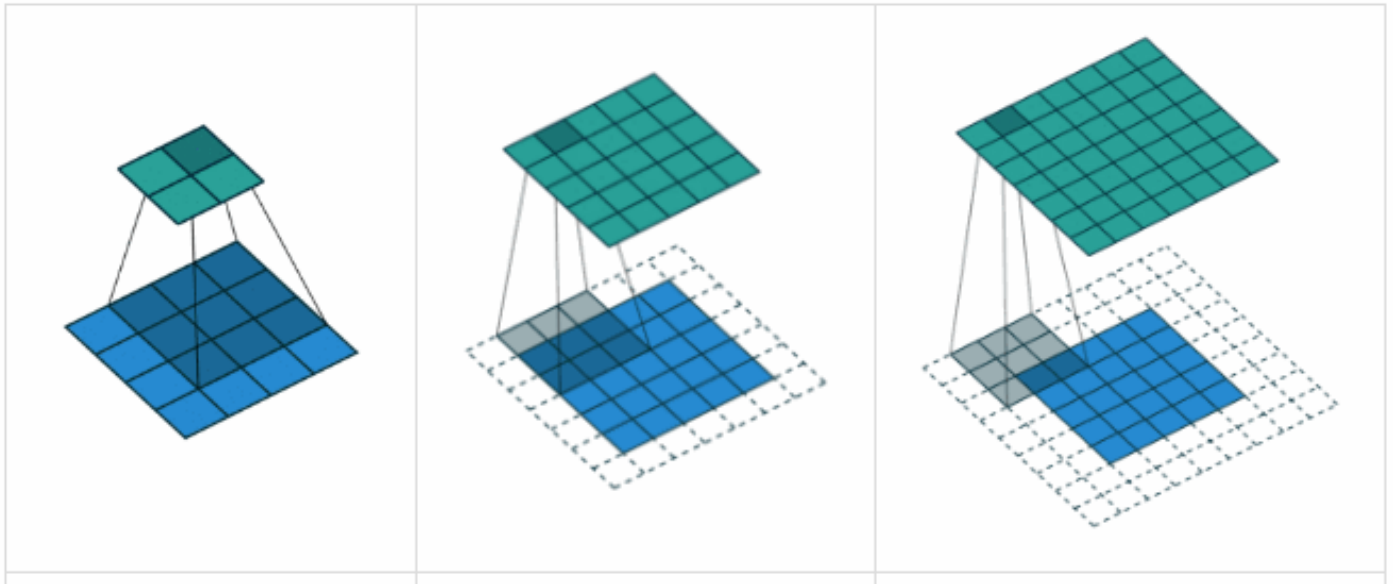
2D Convolution



2D Convolution Boundaries



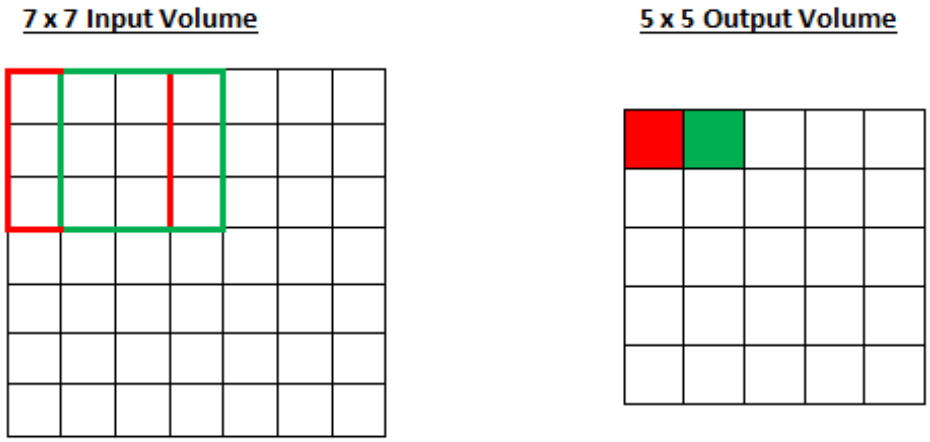
Padding in 2D



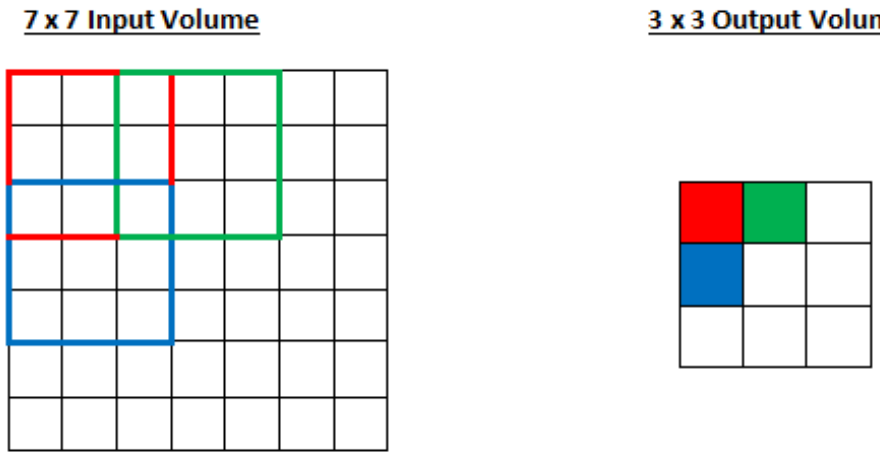
Valid Same Full

Stride – step size

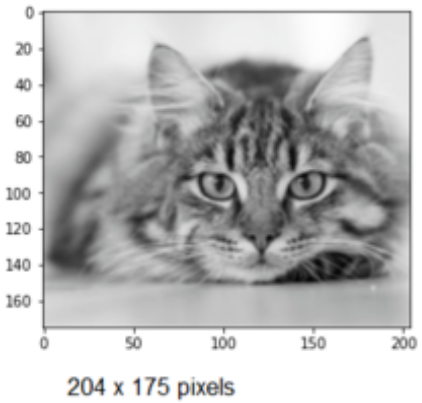
Stride = 1



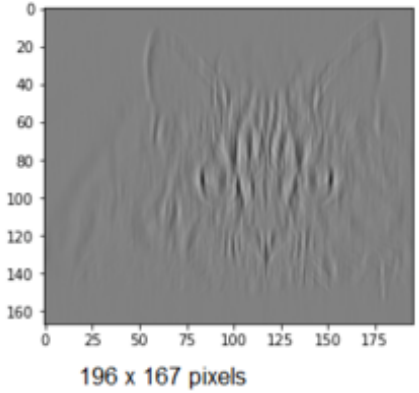
Stride = 2



Convolution Example



$$*\begin{bmatrix} 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \end{bmatrix} =$$



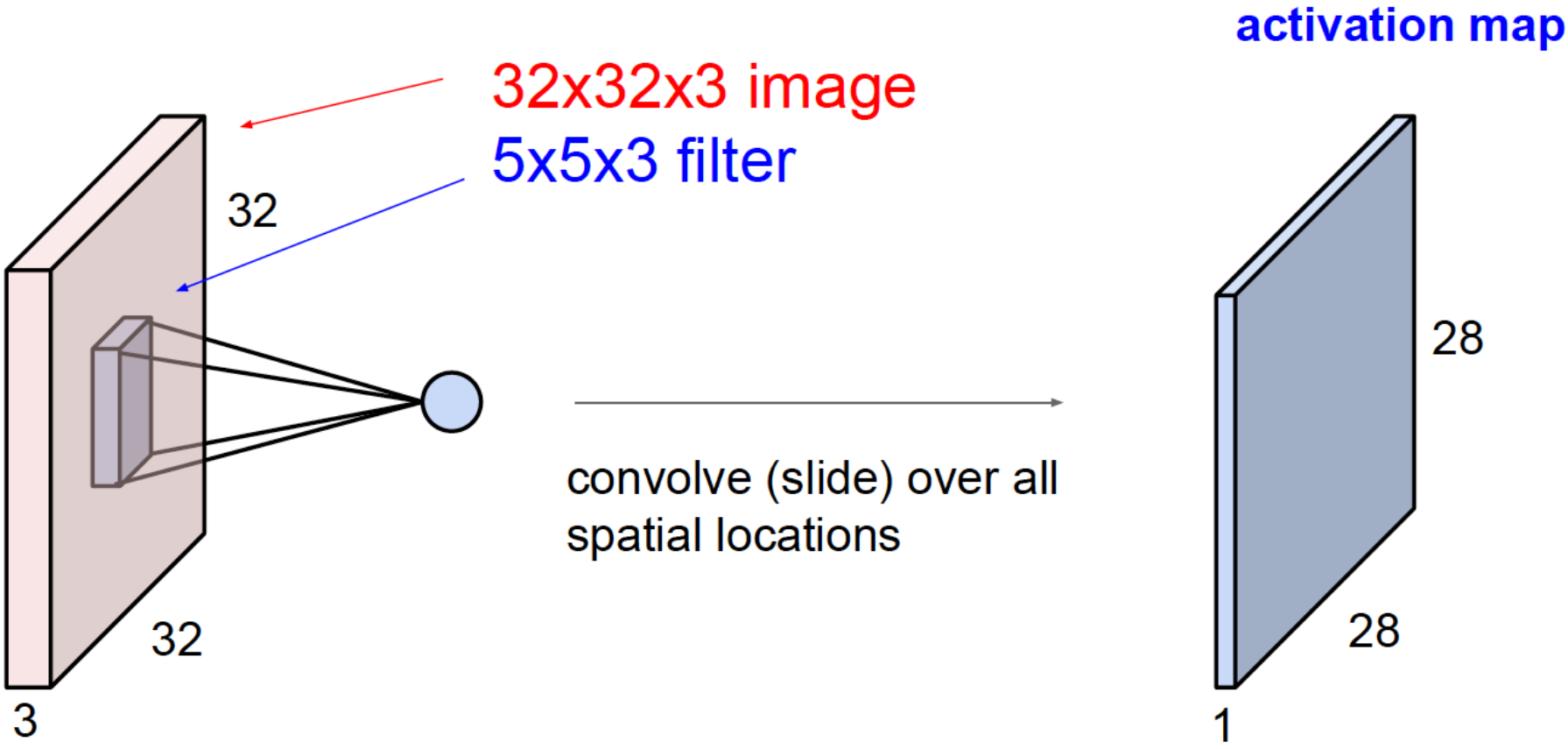
What is this kernel doing?

What does this kernel accomplish?

$$M = \frac{1}{273} \times$$

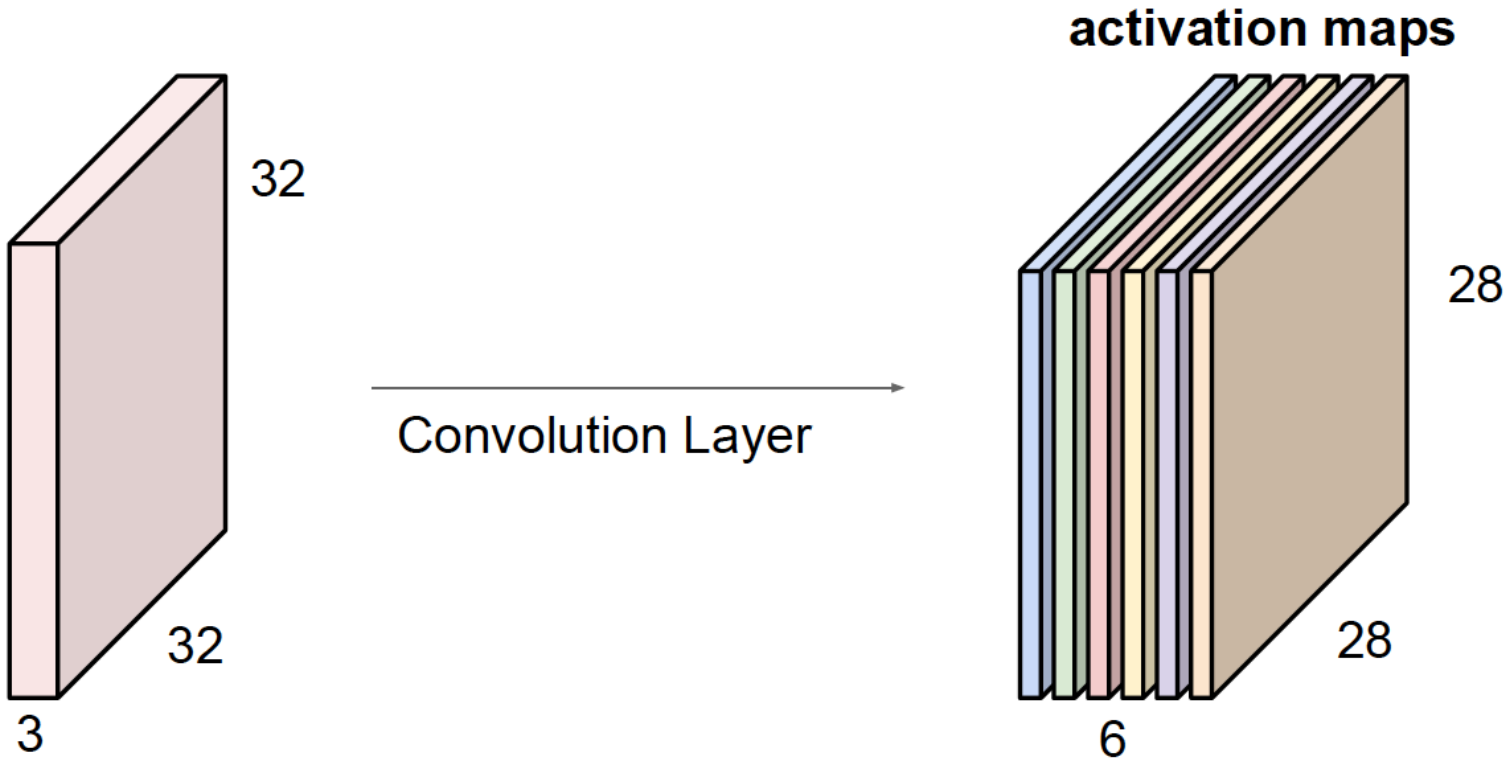
1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1

Convolutional Layers



Convolutional Layers

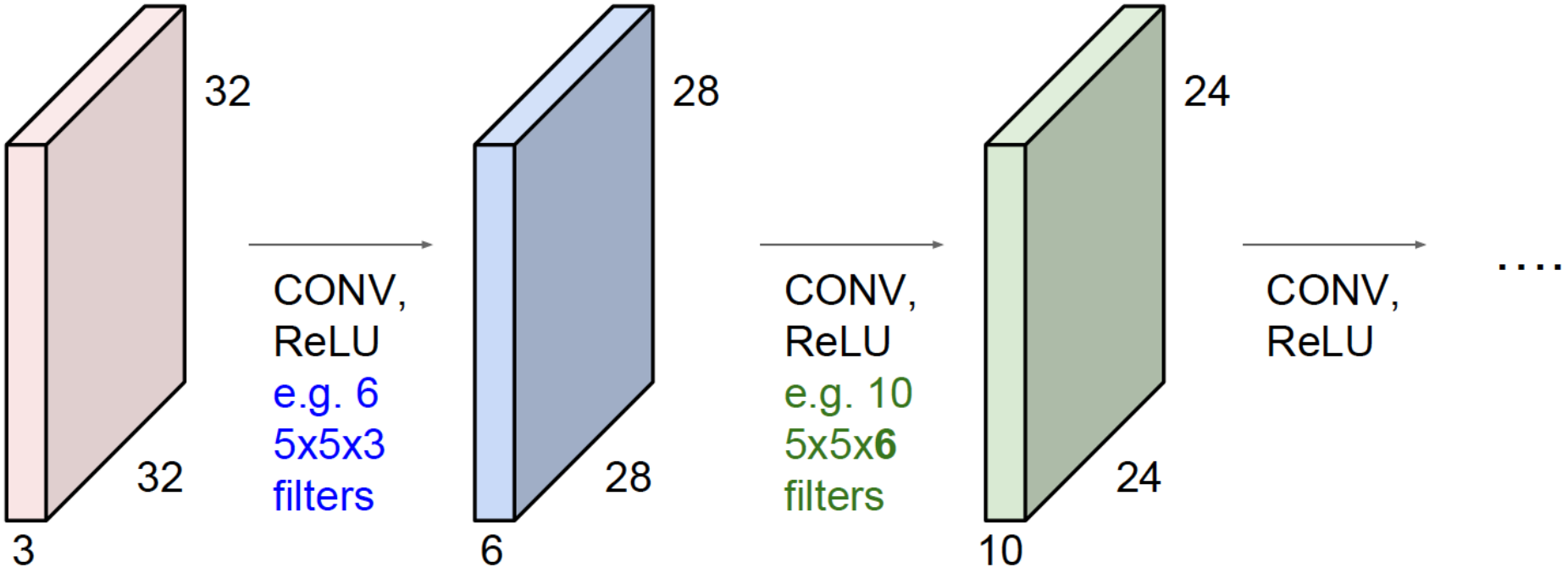
For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:



We stack these up to get a "new image" of size 28x28x6!

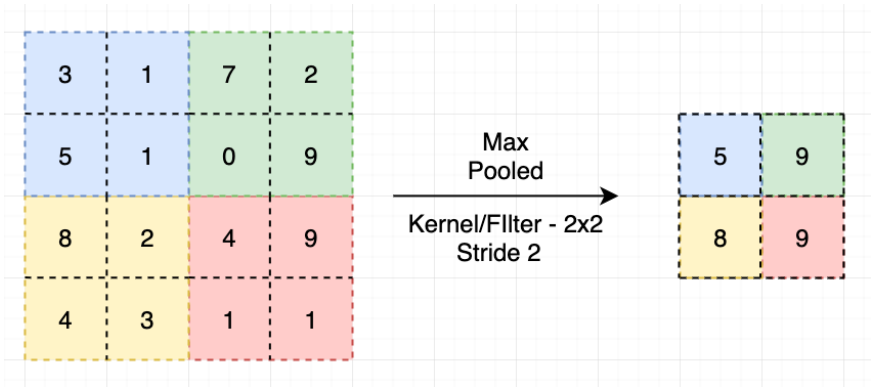
Conv Net

Preview: ConvNet is a sequence of Convolutional Layers, interspersed with activation functions

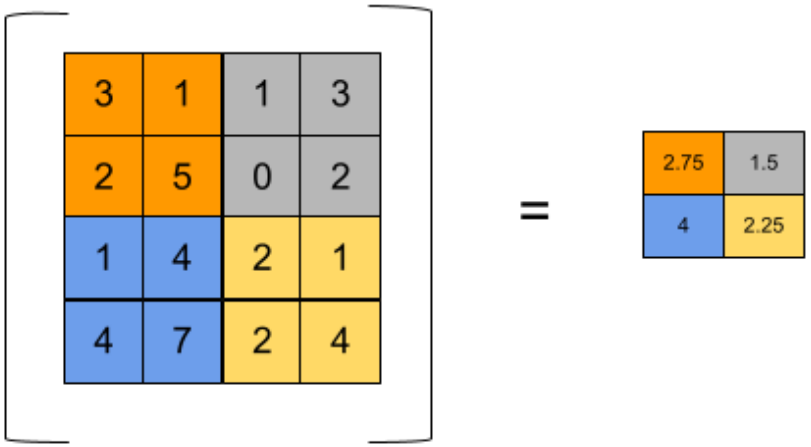


Pooling

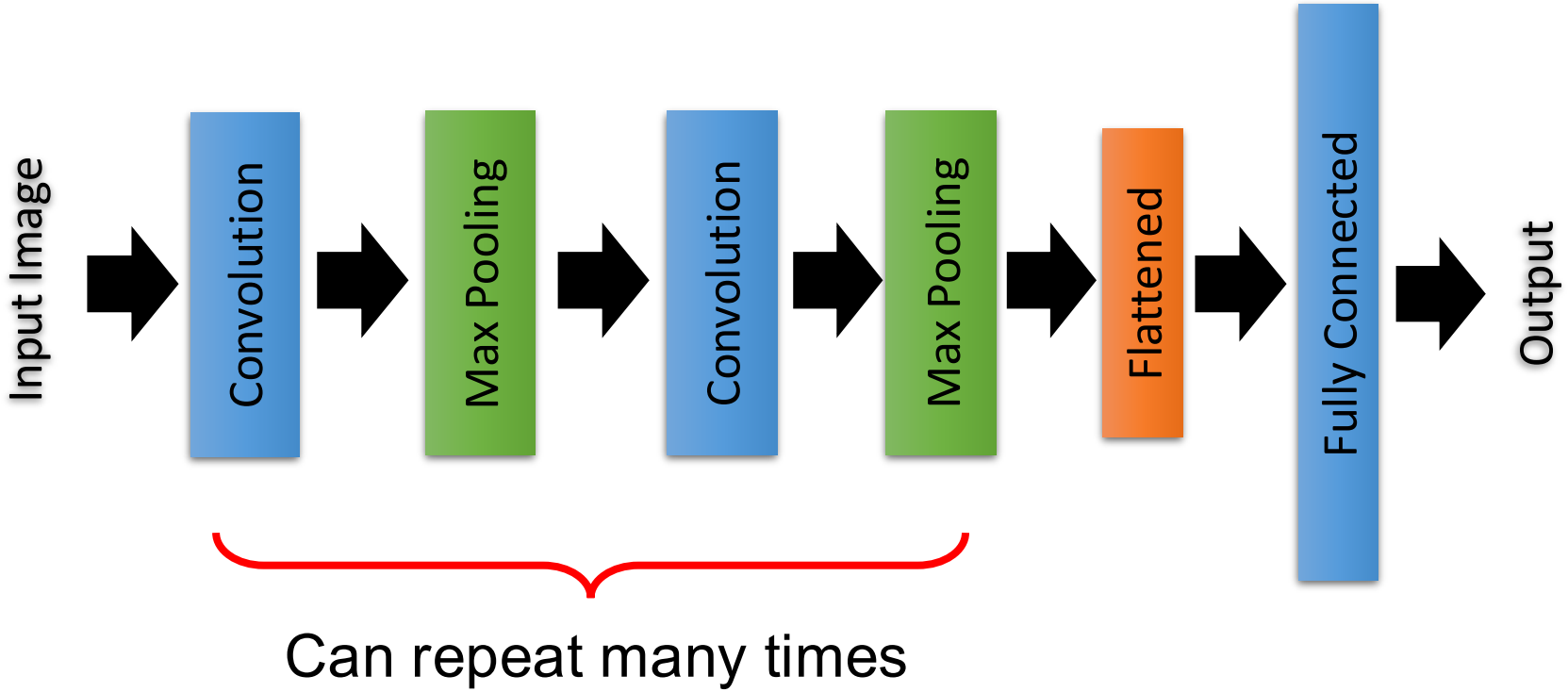
Max Pooling



Avg Pooling



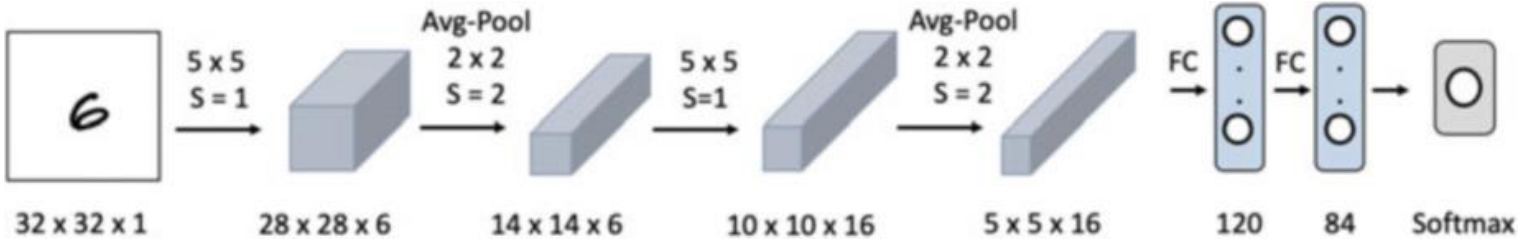
Ingredients of a CNN



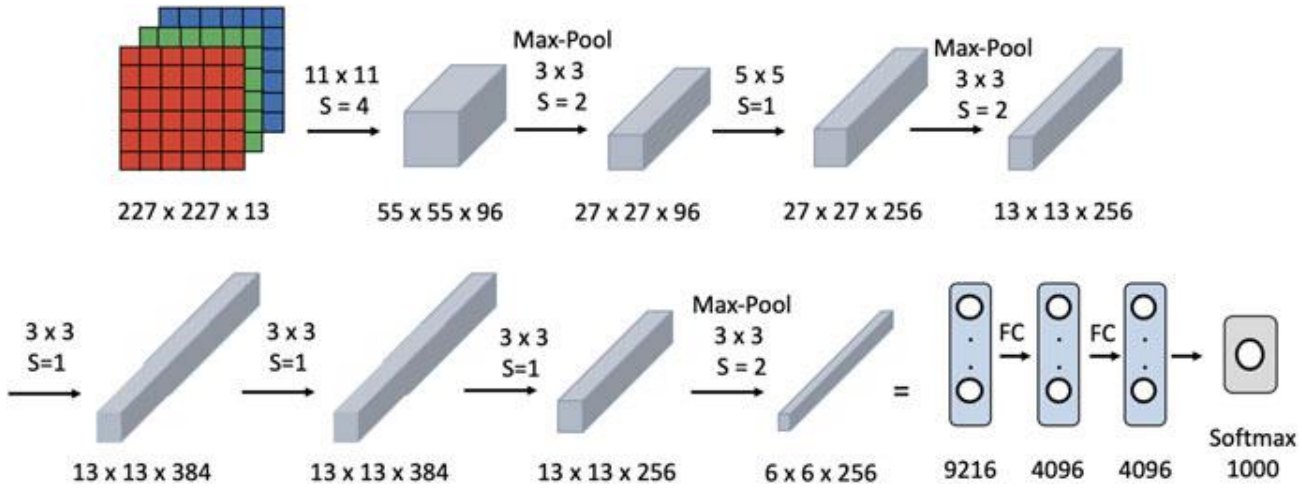
Some Popular Architectures

LeNet and AlexNet

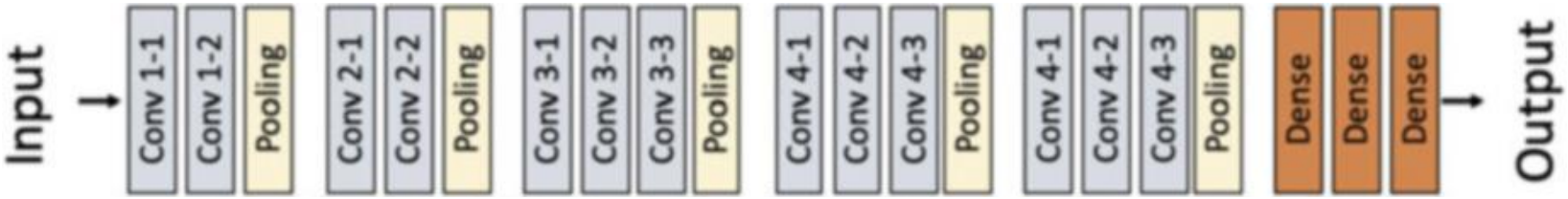
LeNet (1998) – handwritten digit recognition



AlexNet (2012) – ImageNet classification

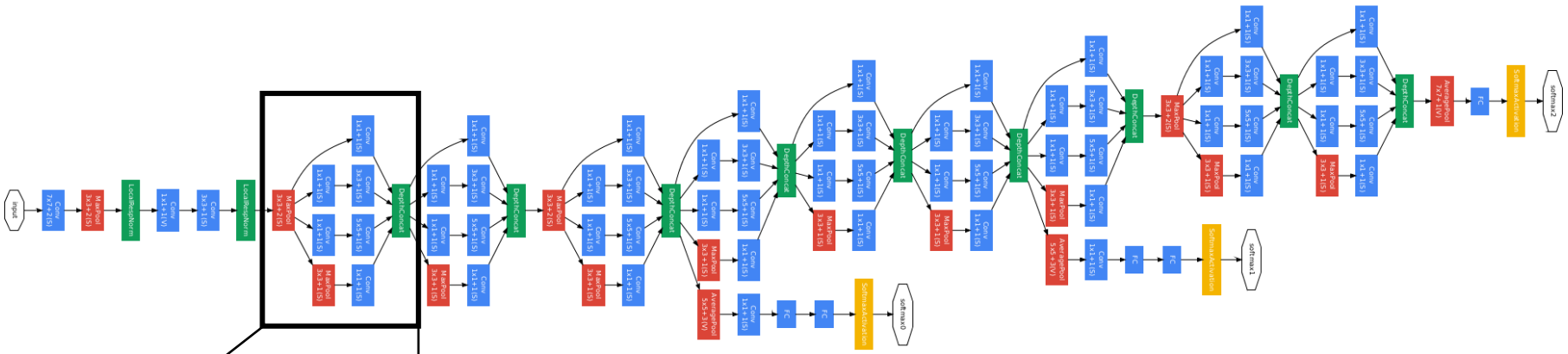


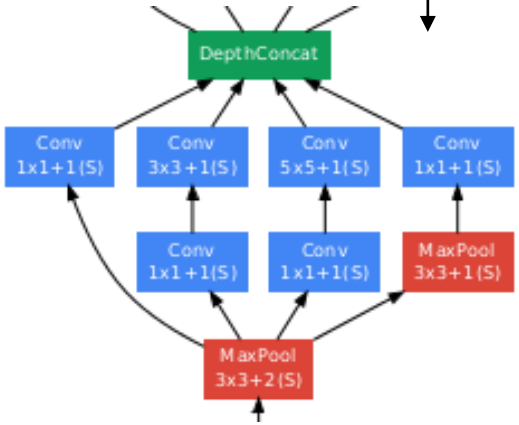
VGG



- A more principled deeper architecture
- Convolutional stages instead of layers
- Standardize filter and pooling choices
  - Convolutions - 3x3 stride 1 pad 1
  - Max pool - 2x2 stride 2

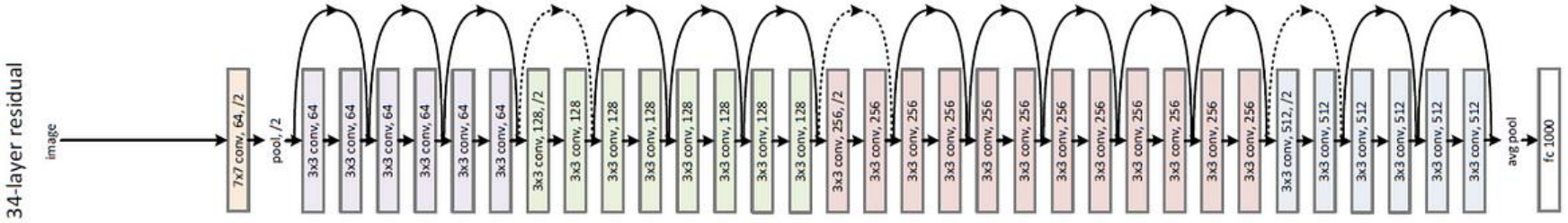
GoogLeNet - Inception





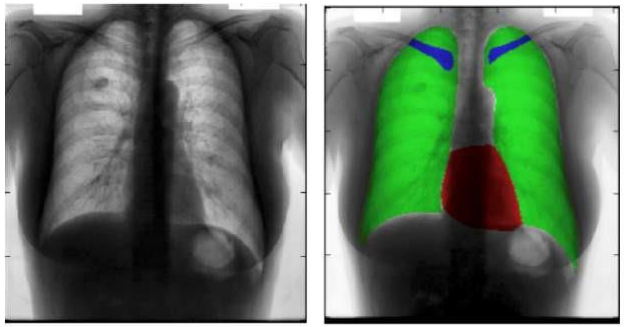
- Use multiple types of filters
- Repeat inception modules
- Initial down sampling
- A more efficient architecture than VGG
- 1x1 convolution

Residual Network Architecture

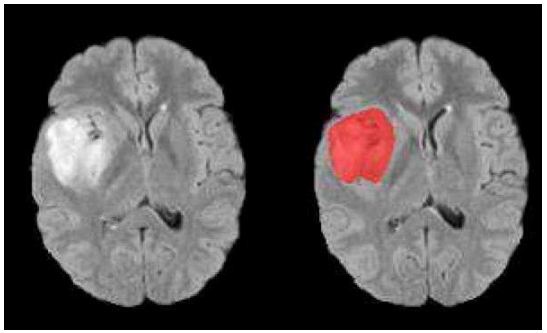


- Skip connections to learn identity functions
  - Makes it easy to have deeper architectures

U-Net Architecture for Medical Image Segmentation



Chest X-Ray

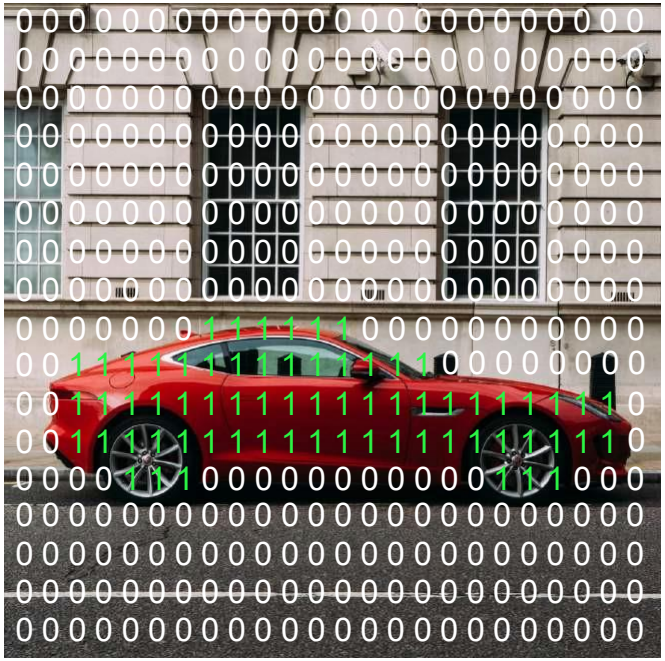


Brain MRI

[Novikov et al., 2017, Fully Convolutional Architectures for Multi-Class Segmentation in Chest Radiographs]
[Dong et al., 2017, Automatic Brain Tumor Detection and Segmentation Using U-Net Based Fully Convolutional Networks]

Credit:
Andrew Ng

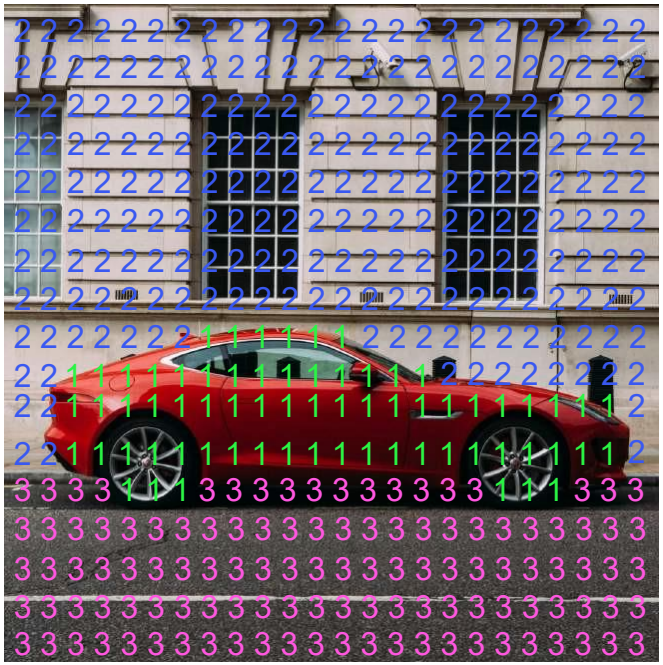
Per-pixel class labels



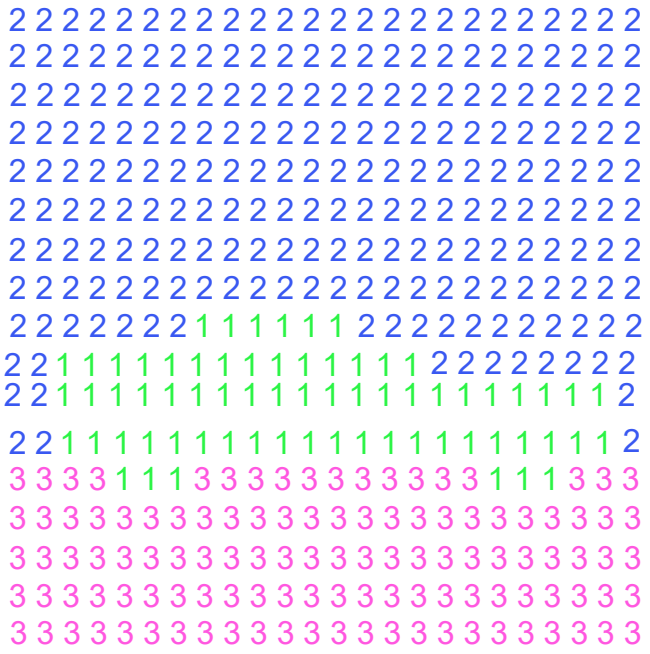
1. Car
0. Not Car

Credit:
Andrew Ng

Per-pixel class labels



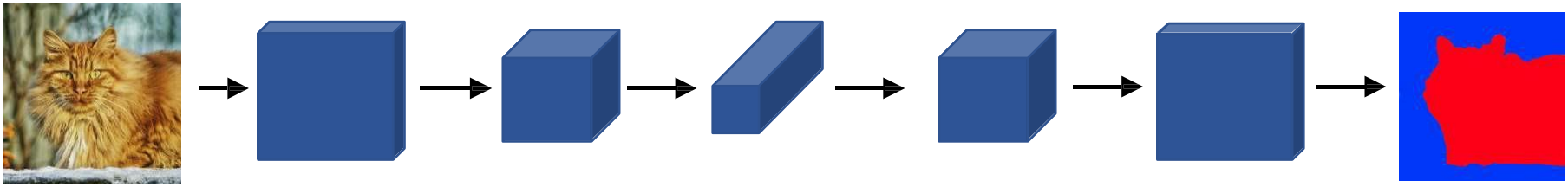
1. Car
2. Building
3. Road



Segmentation Map

Credit:
Andrew Ng

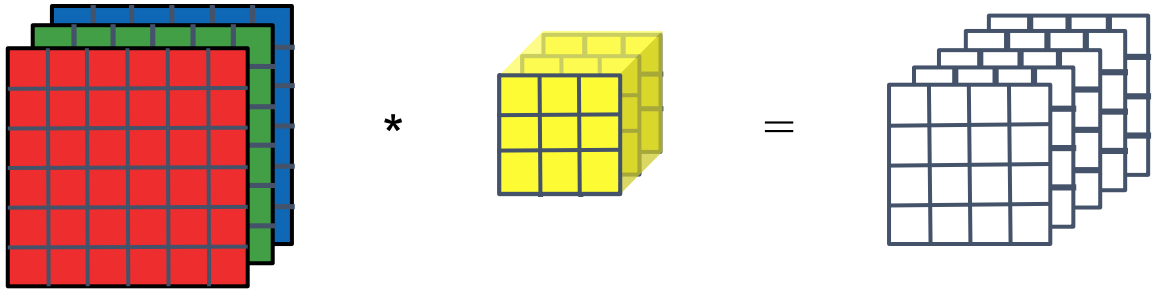
Deep Learning for Semantic Segmentation



Credit:
Andrew Ng

Transpose Convolution

Normal Convolution

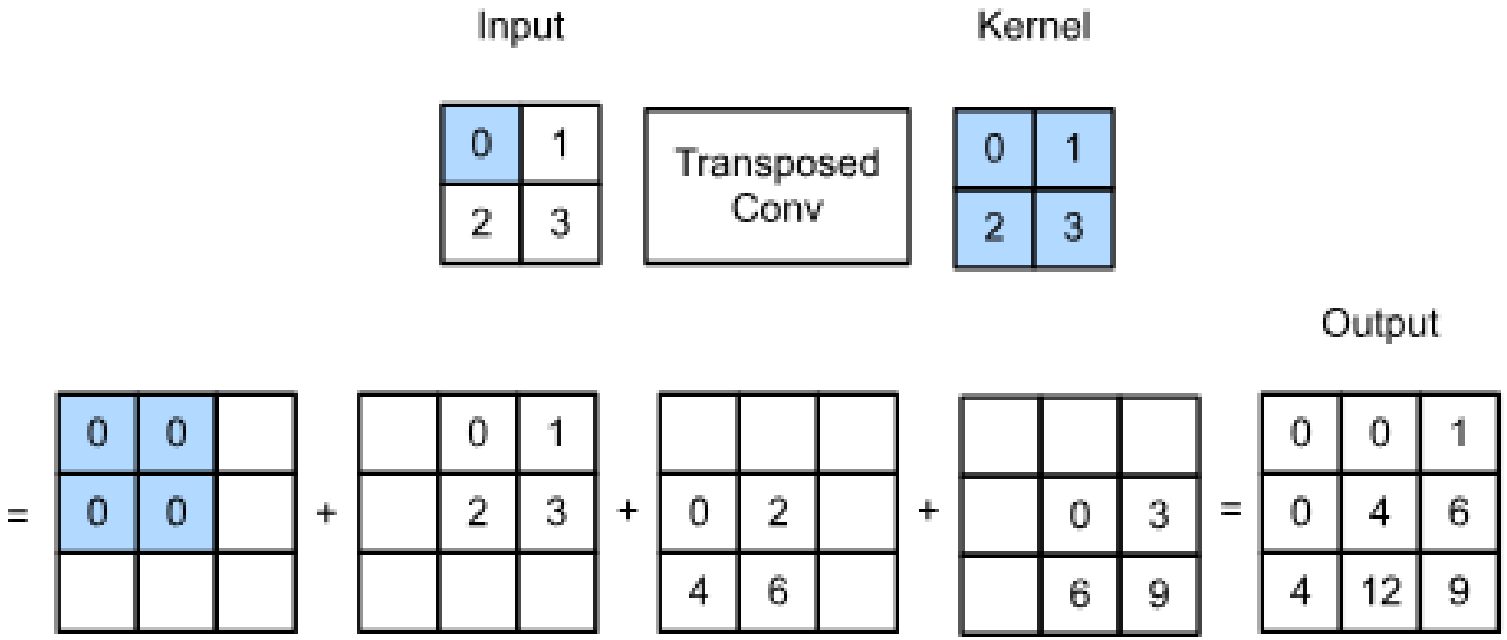


Transpose Convolution

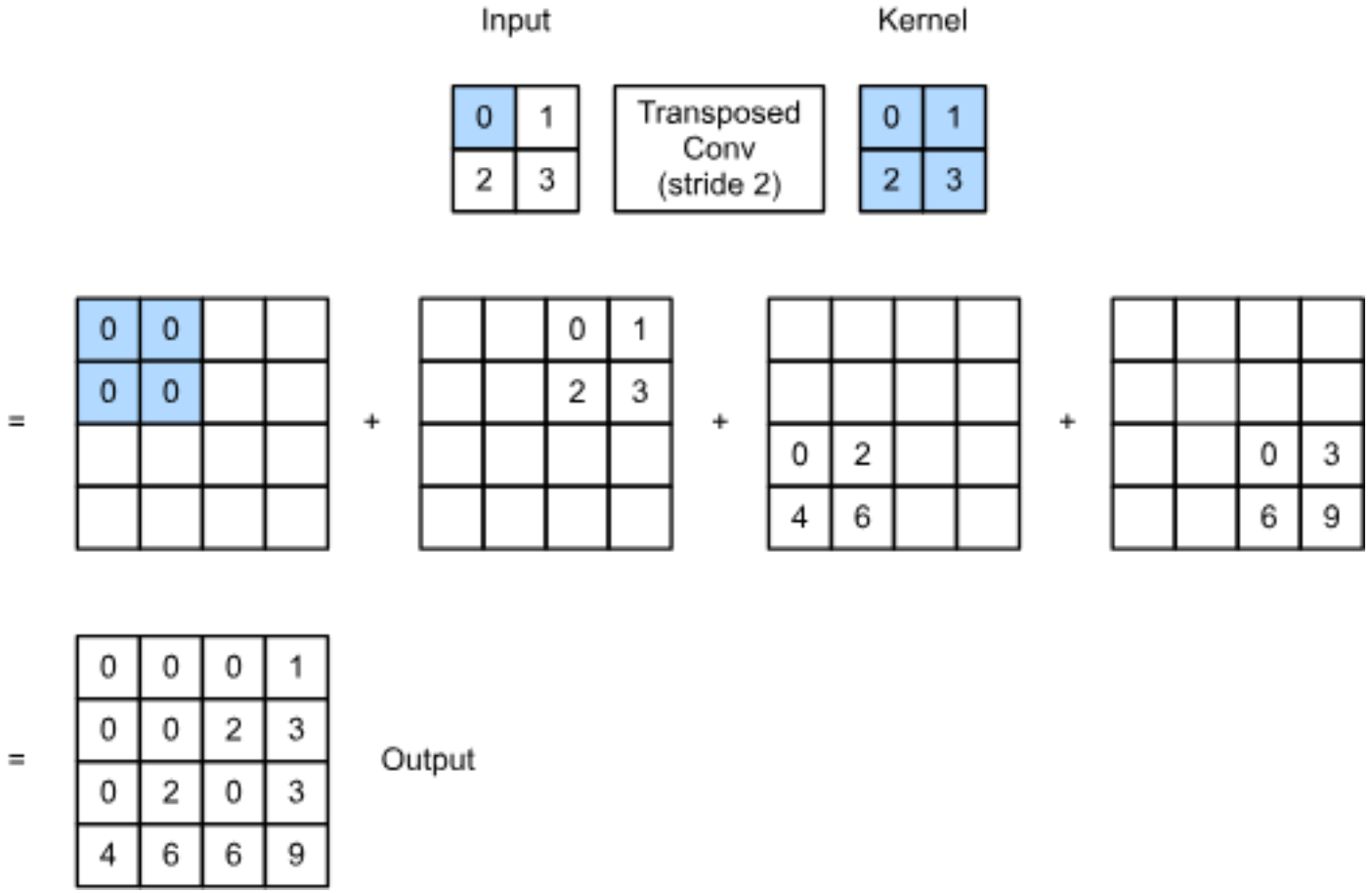


Credit:
Andrew Ng

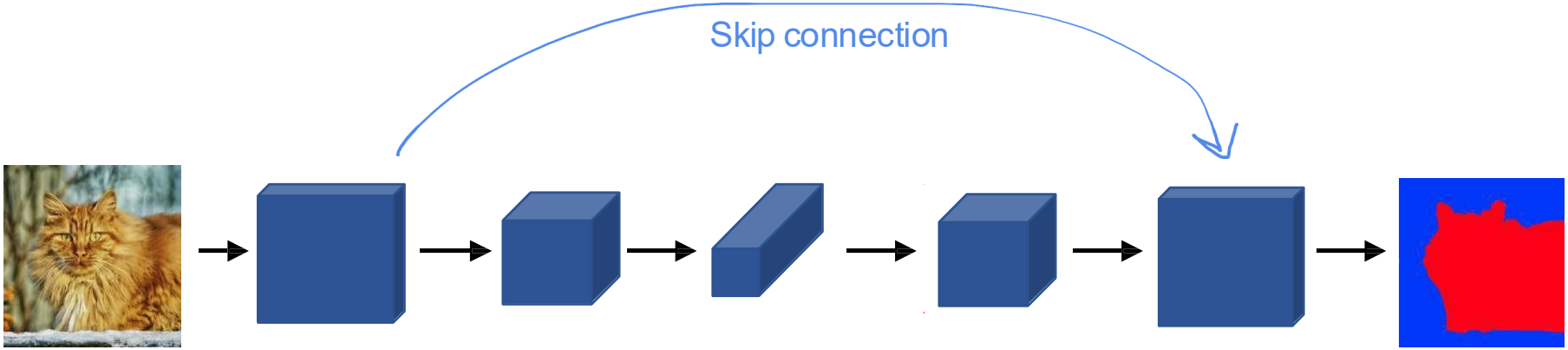
Transpose Convolution



Transpose Convolution

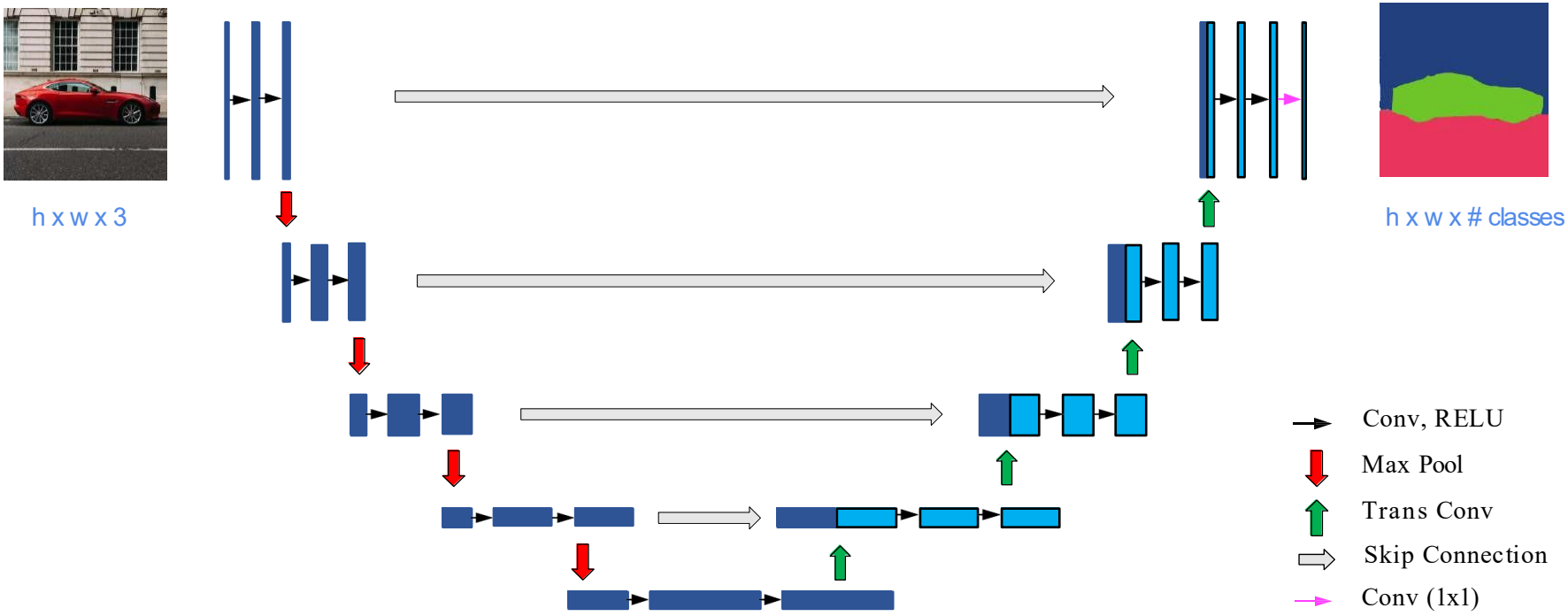


Deep Learning for Semantic Segmentation



Credit:
Andrew Ng

U-Net



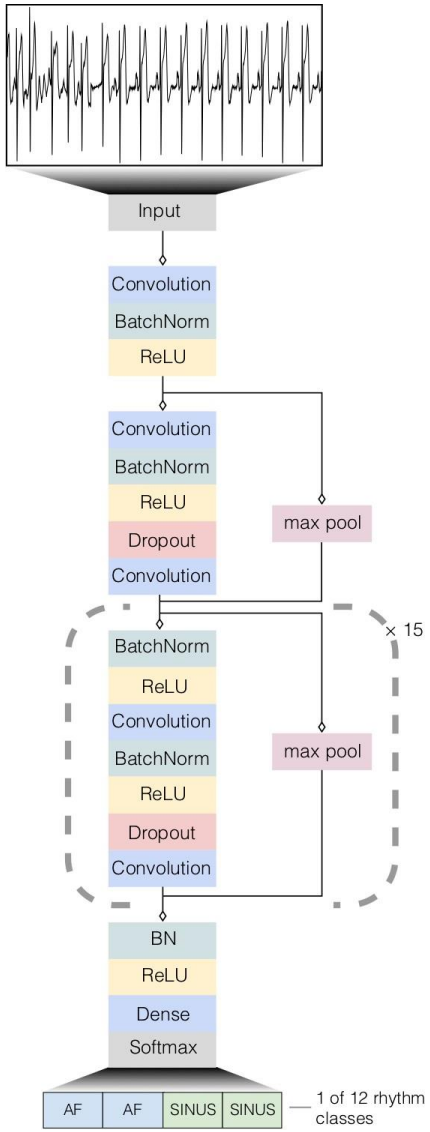
[Ronneberger et al., 2015, U-Net: Convolutional Networks for Biomedical Image Segmentation]

Credit: Andrew Ng

Some Time-series Architectures

ECG Classification

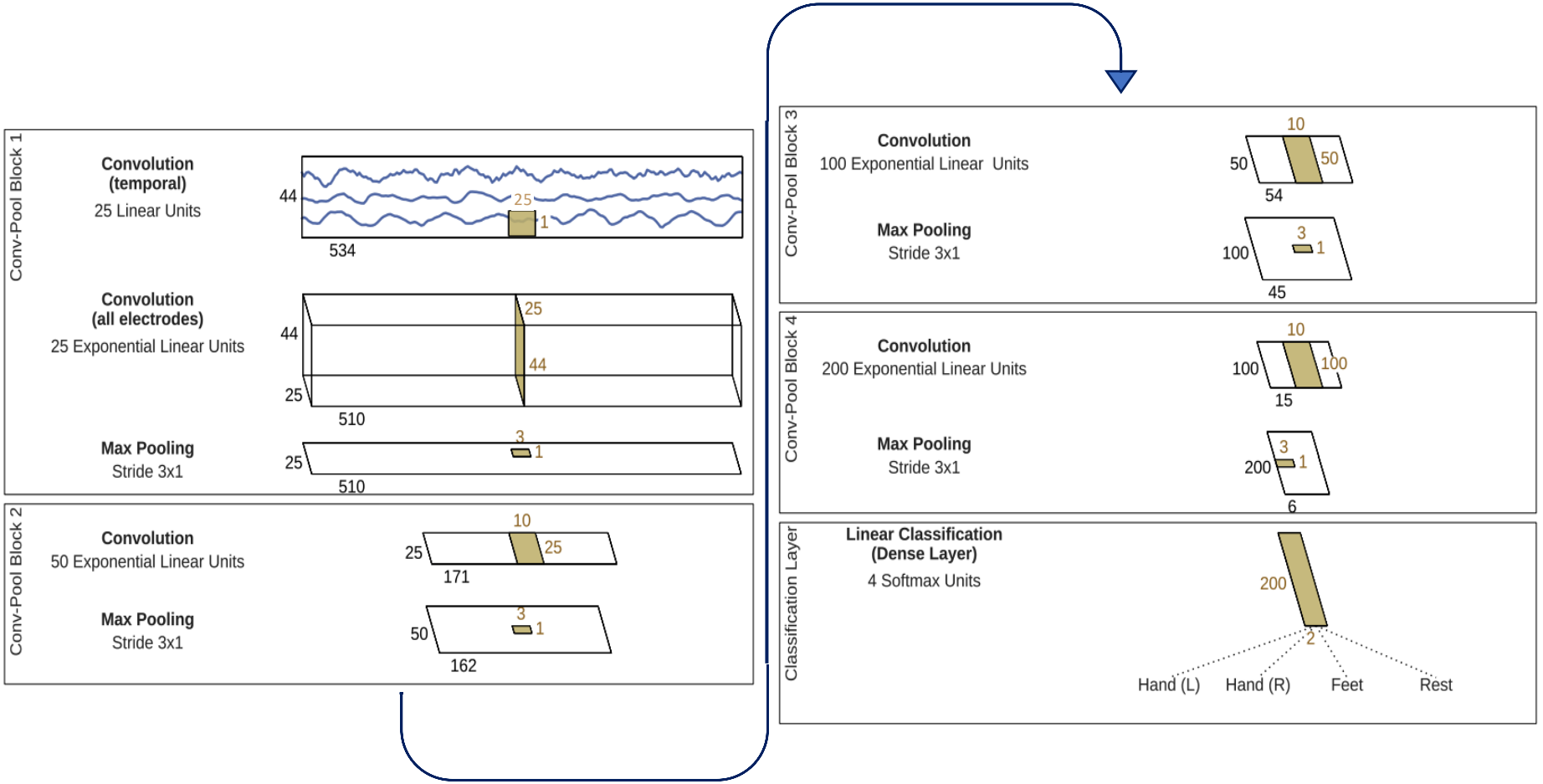
- 30s windows as input (sampled at 200Hz)
- 16 residual blocks with 2 conv layers per block.
- The conv layers all have a filter length of 16 and have 64k filters, where k starts out as 1 and is incremented every 4-th residual block.
- Every alternate residual block subsamples its inputs by a factor of 2 (i.e., max pool).
- When a residual block subsamples the input, the corresponding shortcut connections also subsample their input using a max pooling operation with the same subsample factor.



Source: Nature

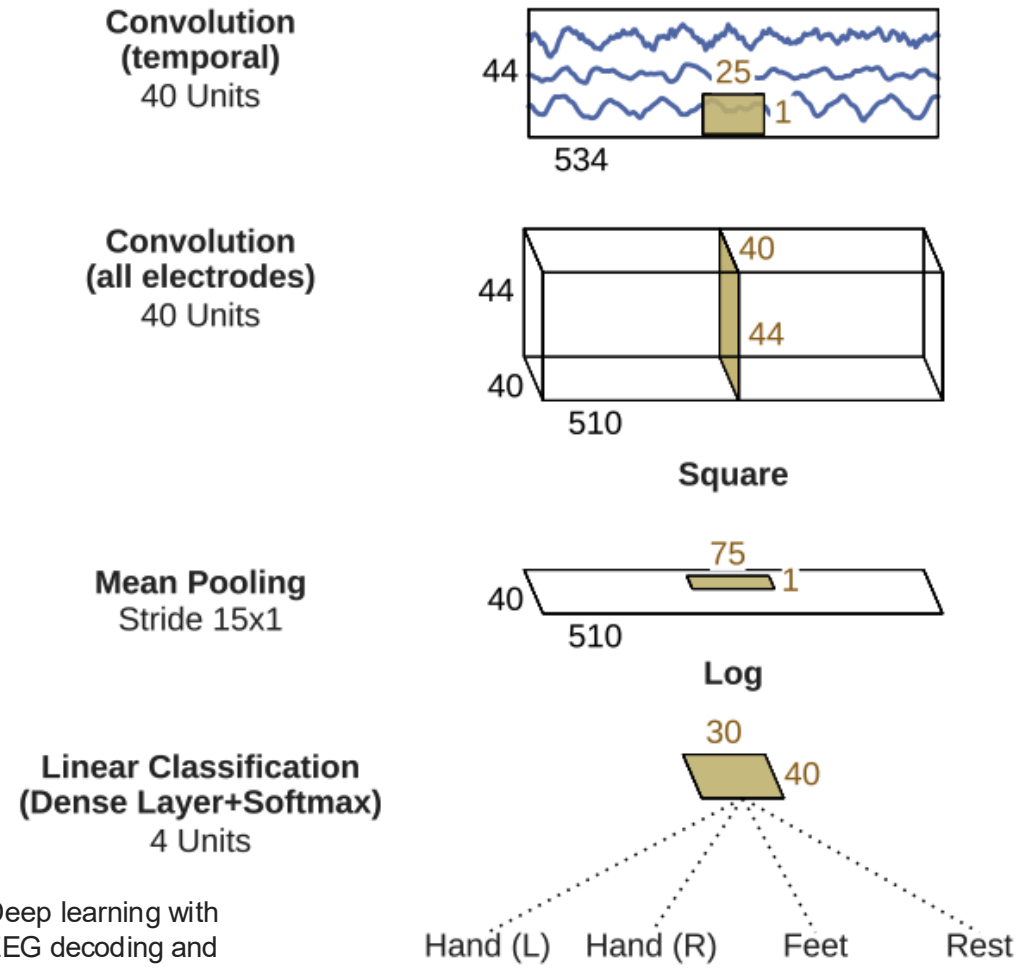
Hannun, A.Y., Rajpurkar, P., Haghpanahi, M. *et al.* Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. *Nat Med*

Multi-channel EEG Classification



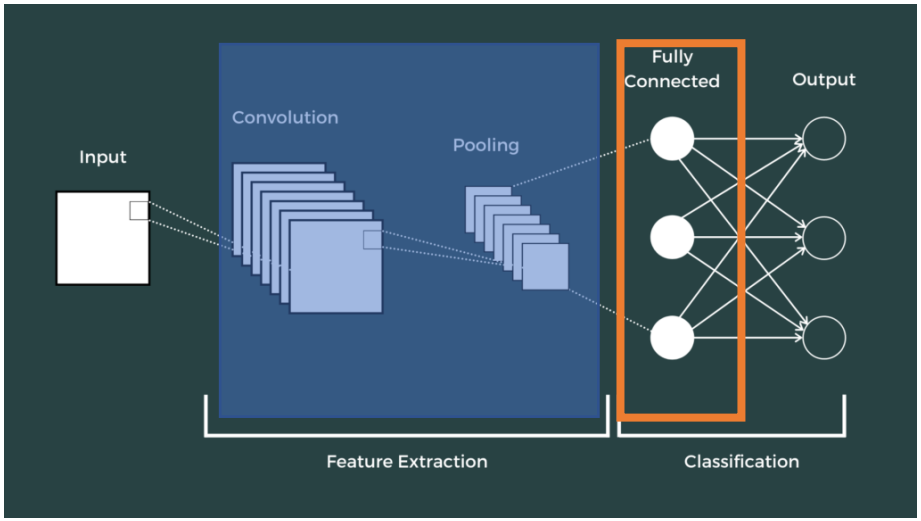
Schirrmeister, Robin Tibor, et al. "Deep learning with convolutional neural networks for EEG decoding and visualization." *Human brain mapping*.

Multi-channel EEG Classification



Schirrmeister, Robin Tibor, et al. "Deep learning with convolutional neural networks for EEG decoding and visualization." *Human brain mapping*.

Transfer Learning



- 1. Freeze the convolutional layers
- 2. Retrain only the fully-connected classifier

Advantage – Use someone else's working solution, often trained on more data

Disadvantage – Locks you into a particular model with no room for improvements or tweaks

Exit pass

https://z.umn.edu/ml4h_exitpass

